Extract — Precision Analysis

Note. This document is an extract from the full dissertation report. It reproduces one analytical section in isolation so that it can be read independently of the rest of the dissertation.

Executive Summary

This extract presents an analytical study of the precision limits of the cup-to-disc ratio, CDR, when segmentation masks are represented on discrete pixel grids and undergo cropping and resizing before being processed by a U-Net. The section derives worst-case and root-mean-square error bounds for quantisation-induced error, shows how these bounds scale with image downsampling, and validates the resulting expressions empirically using the ORIGA dataset.

This section has been selected for standalone presentation because it captures a technically distinctive contribution of the dissertation, not present in current literature: rather than evaluating segmentation performance only through pixel-level metrics such as DICE, it investigates how preprocessing choices impose a theoretical lower bound on the precision of clinically relevant downstream markers.

Crucially, this lower bound is model-invariant: it arises purely from the discretisation and preprocessing pipeline, and therefore imposes a fundamental performance ceiling on the precision of CDR estimation, irrespective of how well a model performs under conventional segmentation metrics such as DICE.

CDR Precision

The accurate measurement of ocular markers, such as CDR, is crucial in the glaucoma screening process. As the mask and image are stored on a pixel grid, CDR can change only in discrete steps. These steps set a theoretical lower bound on the marker's precision. In a clinical setting, it is not sufficient to achieve high pixel-level metrics, as they neglect the downstream ratio errors. Pixel-level metrics, such as DICE, only exist within the resolution of the discretised space, capturing only whole pixels correctly classified but not considering the subsequent error in calculating real-valued markers.

While some papers that utilise U-Nets acknowledge the impact of resizing an image to fit into their model architecture on the precision of marker extraction—and others, which do not, may inherently consider it when designing their architecture—none of the surveyed works in Appendix A.3 actually quantify these effects. This section investigates the effects of preprocessing techniques such as cropping and resizing on CDR precision. Specifically, the aim is to analytically establish

bounds on the maximum and RMS error induced through cropping and resizing (referred to together as "quantisation" for simplicity). Through these bounds, recommendations are provided for crop and model input size settings that guarantee an upper limit of quantisation-induced error; or conversely, given fixed crop-input settings, these bounds offer a clear insight on the maximum and expected induced error and therefore the maximum precision achievable. Following the analytical derivation, the equations are validated against the ORIGA dataset.

It should be noted that these bounds reflect the induced error due to quantisation only, not the model segmentation error. They can be interpreted as a theoretical limit on the precision that can be guaranteed with a perfect classifier. A limitation of this investigation is that it does not consider the sub-pixel effects introduced by post-processing techniques like ellipse-fitting on the predicted mask. In that case, the segmented region is described with an analytical expression rather than a discrete grid; uncertainty propagation would require further analysis of the specific ellipse-fitting method, which is a point for future work.

Boundary Error

Any time a continuous contour is sampled onto a discrete grid of pixels, either by the camera sensor at acquisition *or later by a nearest-neighbour resize* , the sampling operator assigns each contour point to the nearest pixel. Under this rule, the positional error is bounded by $\delta \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ px, i.e. the contour may shift at most half a pixel in either direction. This $\pm\frac{1}{2}$ px bound is therefore the natural starting point for the current analysis; in subsequent subsections, the same bound, rescaled by the downsampling factor B, governs the quantisation error introduced by nearest-neighbour resizing.[1]

Assuming the true boundary position inside a pixel is uniformly distributed, $\delta \sim \mathcal{U}\left[-\frac{1}{2}, \frac{1}{2}\right]$, the variance V_δ and standard deviation σ_δ are

$$V_\delta = \mathrm{Var}[\delta] = \frac{\left(\frac{1}{2} - \left(-\frac{1}{2}\right)\right)^2}{12} = \frac{1}{12}. \tag{2.28}$$

$$\sigma_\delta = \sqrt{\mathrm{Var}[\delta]} = \frac{1}{\sqrt{12}}. \tag{2.29}$$

Propagating Boundary Error to CDR Error

This subsection illustrates the derivation with CDR_v, as defined in Eq. 2.1, however the same algebra applies to CDR_h or RDR. By standard first-order error propagation,

$$\Delta\mathrm{CDR} = \frac{\partial \mathrm{CDR}}{\partial h_{\mathrm{cup}}} \Delta h_{\mathrm{cup}} + \frac{\partial \mathrm{CDR}}{\partial h_{\mathrm{disc}}} \Delta h_{\mathrm{disc}}. \tag{2.30}$$

where

$$\frac{\partial \mathrm{CDR}}{\partial h_{\mathrm{cup}}} = \frac{1}{h_{\mathrm{disc}}}, \qquad \frac{\partial \mathrm{CDR}}{\partial h_{\mathrm{disc}}} = -\frac{h_{\mathrm{cup}}}{h_{\mathrm{disc}}^2}. \tag{2.31}$$

[1] Popular implementations such as `tf.image.resize(..., method='nearest')` follow this round-to-centre rule. If an implementation instead always floors or ceils the coordinate, the bound would double to $|\delta| = 1$ px; this conservative alternative yields $\sigma_\delta = 1/\sqrt{3}$, and the rest of the analysis is scaled accordingly.

Here, Δh_{cup} and Δh_{disc} are the total vertical errors in the cup-height and disc-height measurements, respectively, introduced by the positional errors of the two boundaries of each structure. Concretely, if δ_{bot} and δ_{top} are the errors of the bottom and top structure boundaries, then

$$\Delta h_{\text{cup}} = \delta_{\text{bot,cup}} - \delta_{\text{top,cup}}, \qquad \Delta h_{\text{disc}} = \delta_{\text{bot,disc}} - \delta_{\text{top,disc}}. \tag{2.32}$$

Since each boundary error δ has variance V_δ and the top and bottom errors are independent, it follows that

$$\text{Var}[\Delta h_{\text{cup}}] = \text{Var}[\delta_{\text{bot,cup}}] + \text{Var}[\delta_{\text{top,cup}}] = \frac{2}{12}, \tag{2.33}$$

$$\text{Var}[\Delta h_{\text{disc}}] = \text{Var}[\delta_{\text{bot,disc}}] + \text{Var}[\delta_{\text{top,disc}}] = \frac{2}{12}. \tag{2.34}$$

Absolute Worst Case. In this case, all four boundaries of the cup and disc shift in lockstep so that the signed error attains its maximum magnitude. For instance, if the cup's bottom and top edges each shift outward by $+\frac{1}{2}$ px and $-\frac{1}{2}$ px respectively, then

$$\Delta h_{\text{cup}} = \left(+\tfrac{1}{2}\right) - \left(-\tfrac{1}{2}\right) = +1 \text{ px}, \qquad \Delta h_{\text{disc}} = \left(-\tfrac{1}{2}\right) - \left(+\tfrac{1}{2}\right) = -1 \text{ px}, \tag{2.35}$$

or vice versa. Hence

$$\Delta h_{\text{cup}} = \pm 1\,\text{px}, \qquad \Delta h_{\text{disc}} = \mp 1\,\text{px}, \tag{2.36}$$

and the induced CDR error is

$$\Delta\text{CDR}_{\text{max}} = \pm\frac{1}{h_{\text{disc}}} \pm \frac{h_{\text{cup}}}{h_{\text{disc}}^2} = \pm\frac{1+\text{CDR}}{h_{\text{disc}}}. \tag{2.37}$$

Approximate Worst Case. A more relaxed approximation neglects the contribution of Δh_{disc}, since it is multiplied by $\frac{h_{\text{cup}}}{h_{\text{disc}}^2}$ and is therefore less significant. Setting $\Delta h_{\text{disc}} = 0$ and $\Delta h_{\text{cup}} = \pm 1$ gives

$$\overline{\Delta\text{CDR}} = \pm\frac{1}{h_{\text{disc}}}. \tag{2.38}$$

Root-Mean-Square (RMS) Error. Unlike the worst-case bound, $\Delta\text{CDR}_{\text{max}}$, the RMS error σ_{CDR} captures the typical size of the quantisation error. By linear error propagation, using the variance expressions in Eqs. (2.33) and (2.34),

$$\begin{aligned}\sigma_{\text{CDR}} &= \sqrt{\left(\frac{1}{h_{\text{disc}}}\right)^2 \frac{2}{12} + \left(-\frac{h_{\text{cup}}}{h_{\text{disc}}^2}\right)^2 \frac{2}{12}} \\ &= \frac{\sqrt{1+\text{CDR}^2}}{h_{\text{disc}}\sqrt{6}}.\end{aligned} \tag{2.39}$$

Approximate RMS Error. As before, if the disc contribution is neglected in a more relaxed approximation, only Δh_{cup} remains. Hence

$$\bar{\sigma}_{\text{CDR}} = \frac{1}{h_{\text{disc}}\sqrt{6}}. \tag{2.40}$$

Quantisation Blocks

To keep computation tractable, the U-Net requires full-resolution images to be downsampled into much smaller dimensions for feasible input sizes. When downsampling via nearest-neighbour, the original fine grid is replaced by a coarser one: every geometric length is measured in larger coarse-pixel units. In other words, a segment spanning L pixels at full resolution spans L/B pixels after the image is reduced by the downsampling factor B. Equivalently, one coarse-pixel step corresponds to B full-resolution pixels. Intuitively, cropping before downsampling reduces information loss: cropping alone does not change resolution; it merely reduces the dimensions of the image by cutting away context. Therefore, cropping before downsampling reduces B, defined as

$$B = \frac{H_{\text{crop}}}{H_{\text{input}}}. \tag{2.41}$$

where H_{crop} (px) is the height of the cropped region of interest and H_{input} (px) is the height of the downsampled image presented to the U-Net.

If h_{disc} (px) is the optic-disc height in the full-resolution image, its effective height in the downsampled image is

$$h_{\text{disc}}^{\text{eff}} = \frac{h_{\text{disc}}}{B}. \tag{2.42}$$

Since a single coarse-pixel step corresponds to B full pixels, each pixel-level error bound derived earlier must be scaled by a factor of B in any expression that measures height in full-pixel units. Substituting $h_{\text{disc}}^{\text{eff}} = h_{\text{disc}}/B$ into the previous expressions yields

$$_B\Delta\text{CDR}_{\max} = \pm\frac{B(1+\text{CDR})}{h_{\text{disc}}}. \tag{2.43}$$

$$_B\overline{\Delta\text{CDR}} = \pm\frac{B}{h_{\text{disc}}}. \tag{2.44}$$

$$_B\sigma_{\text{CDR}} = \frac{B\sqrt{1+\text{CDR}^2}}{h_{\text{disc}}\sqrt{6}}. \tag{2.45}$$

$$_B\bar{\sigma}_{\text{CDR}} = \frac{B}{h_{\text{disc}}\sqrt{6}}. \tag{2.46}$$

Design Tool

The quantisation bounds derived above can now be turned into a practical design tool for the crop-resize stage of a U-Net pipeline. Since quantisation error is training-invariant, its effect on precision must be considered in the architecture design stage in the context of its use case.

Clinical Tolerance. Clinical grading itself is noisy: [39] found the 95% confidence limit for the mean inter-observer difference in CDR_v is 0.11. Meanwhile, other studies adopt decision steps of 0.05 and treat smaller shifts as clinically irrelevant [40]. Therefore, the design precision is set to $\varepsilon = 0.05$, which matches the increments used in the literature and ensures that a worst-case

quantisation error

$$|_B\Delta\mathrm{CDR}_{\max}| \leq \varepsilon \tag{2.47}$$

cannot push a measurement beyond agreed limits of clinical variability.

Solving for the scale factor. To guarantee $|_B\Delta\mathrm{CDR}_{\max}| \leq \varepsilon$, it is required that

$$B \leq \frac{\varepsilon \times h_{\mathrm{disc}}}{1 + \mathrm{CDR}}. \tag{2.48}$$

Equation (2.48) contains two dataset-dependent quantities, CDR and h_{disc}. Taking the most adverse values observed in the ORIGA training set in the most conservative approach, $\mathrm{CDR}_{v,\max} = 0.96$ and $h_{\mathrm{disc,min}} = 254$ px yields $B \lesssim 6.48$.

Unknown Populations. Acknowledging that in deployment, the true extrema of CDR_v and h_{disc} are unknown, large population studies, such as [3], provide percentile curves for both quantities. Using the 99th percentile for CDR_v and the 1st for h_{disc} widens the safety margin only slightly, the effects of which are neglected here.

Design Choices. $B = H_{\mathrm{crop}}/H_{\mathrm{input}}$ now links two free variables:

- H_{input}: constrained by CPU/GPU memory and U-Net architecture.
- H_{crop}: determined by the ROI detector.

One can set either one to the largest value the pipeline permits, and choose the other so that $B \leq 6.48$. Conversely, if both sizes are predetermined, the same inequality yields the minimum achievable CDR_v precision for that configuration. This study selects $H_{\mathrm{crop}} = 620$, as determined in Section 2.3.5, and the H_{input} of the U-Net model is 256×256. Therefore, in this study $B = 2.5$, limiting the CDR_v precision to approximately 0.02.

Validation

To empirically confirm the error bounds outlined in Eqs. (2.43)–(2.46), the CDR_v values of all samples in the ORIGA dataset are evaluated in full-resolution and in a series of downsampled resolutions. First, as the images are different dimensions, all images undergo a square crop of size $H_{\mathrm{crop}} = 1024\,\mathrm{px}$ centred around the OD centroid.[2] From the cropped images the h_{disc} and h_{cup} are calculated. Each cropped image is resized to

$$H_{\mathrm{input}} \in \{64, 128, 256, 512\},$$

corresponding to $B \in \{16, 8, 4, 2\}$. The $h^{\mathrm{eff}}_{\mathrm{disc}}$ and $h^{\mathrm{eff}}_{\mathrm{cup}}$ are calculated in the downsampled space, the CDR_v is calculated and the difference from the original, $\Delta\mathrm{CDR}$, is recorded.

[2] In the case that the OD centroid lies too close to the image border for such a crop, it is shifted. The OD is no longer perfectly centred but always fully contained.

Across all images the $(h^{\text{eff}}_{\text{disc}}, \Delta\text{CDR})$ pair is recorded separately for the four values of B. The results are presented in Figure 2.3 for $B = 2$ and the rest can be found in Appendix A.8. The interquartile ranges in all cases lie well within the accurate RMS bounds. Almost all whisker endpoints are contained within the approximate worst-case.

As the figure illustrates, for all bins the median ΔCDR remains very close to zero, almost exactly zero for most of the bins. This indicates good agreement with the zero-centred, symmetric model outlined so far.

Table 2.2 reports, for each H_{input}, the percentages of all samples whose ΔCDR falls within the upper and lower analytic bounds. As expected, the accurate worst-case bound captures 100% of values in all cases. Between 94% and 98% of values are successfully captured even by the approximated worst-case bound. The accurate RMS bound successfully captures the majority of cases as expected and the approximated RMS captures just over half the values.

These observations and the overall coverage percentages demonstrate excellent agreement between the empirical quantisation errors and the analytic predictions.

Table 2.2: Percentage of images (across all $h^{\text{eff}}_{\text{disc}}$ bins) whose quantisation error ΔCDR falls within each analytic bound, for each network input size.

Input Size	${}_B\Delta\text{CDR}_{\max}$	${}_B\overline{\Delta\text{CDR}}$	${}_B\sigma_{\text{CDR}}$	${}_B\bar{\sigma}_{\text{CDR}}$
64×64	100.0%	95.08%	62.31%	56.15%
128×128	100.0%	94.31%	60.00%	53.08%
256×256	100.0%	95.54%	63.54%	58.77%
512×512	100.0%	98.31%	63.69%	58.62%

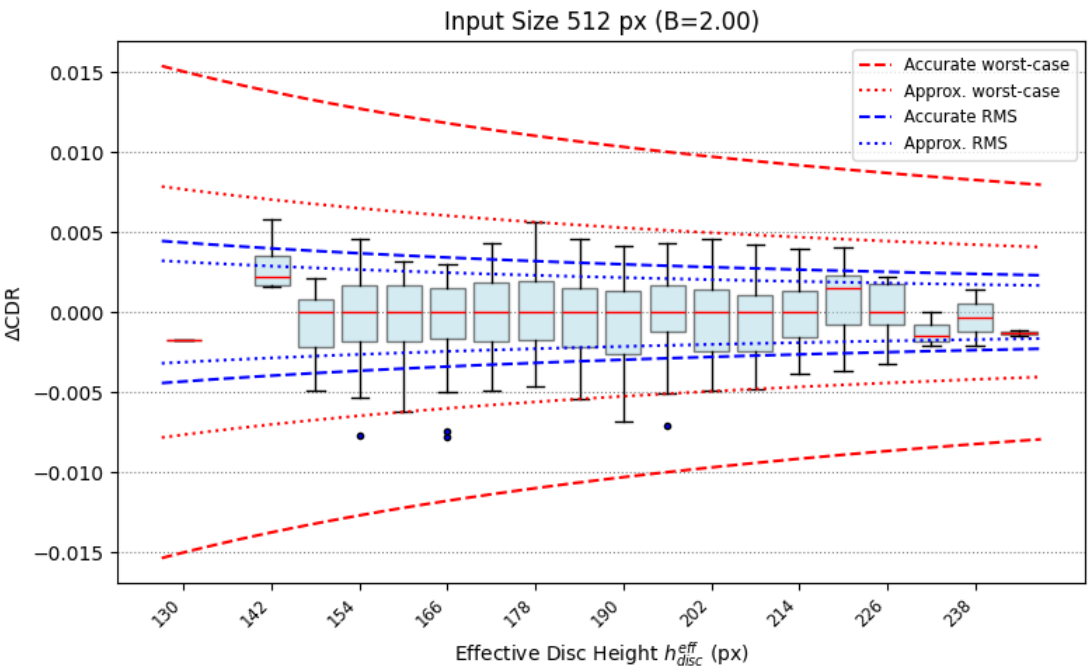

Figure 2.3: The pairs are partitioned into $n = 20$ equal-width bins where the ΔCDR is summarised by its median, 75th and 90th percentiles in box-and-whisker plots. Overlaid are the analytical bounds derived in this extract.

References

[3] Crowston JG, Hopley CR, Healey PR, Lee A, Mitchell P, Blue Mountains Eye Study. The effect of optic disc diameter on vertical cup-to-disc ratio percentiles in a population based cohort: the Blue Mountains Eye Study. *The British Journal of Ophthalmology*. 2004 Jun;88(6):766–770.

[39] Haslett RS, Batterbury M, Cuypers M, Cooper RL. Inter-observer agreement in clinical optic disc measurement using a modified 60 D lens. *Eye*. 1997;11(Pt 5):692–697.

[40] Arnalich-Montiel F, Muñoz-Negrete FJ, Rebolleda G, Sales-Sanz M, Cabarga C. Cup-to-disc ratio: agreement between slit-lamp indirect ophthalmoscopic estimation and Stratus optical coherence tomography measurement. *Eye*. 2007;21(8):1041–1049. Available from: `https://doi.org/10.1038/sj.eye.6702391`.